Social Interaction Breaks Replicate Independence in Controlled-Clone LLM Agents

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Abstract

Multi-agent LLM evaluations and social simulations create many agents from one model and treat them as comparable replicates, even after the agents interact. This assumption has not been tested with matched controls. We test it with SocCogEval, a controlled-clone assay for Social Cognitive Transfer (SCT), and find that it fails. Agents share the same base model, system prompt, and decoding parameters, and a private brief is their only initial difference. A factorial protocol separates the brief, 30 rounds of interaction, and initialization noise, and a matched solo-generation condition bounds the role of extended generation. On Qwen3-122B-FP8, toggling interaction with the briefs held fixed raises off-topic BFI-10 response-profile drift by $\bar{\Delta}$=+0.917 on Greenfield (95% CI [+0.620,+1.205], exact p=0.0008) and +0.912 on Riverton ([+0.584,+1.255], p=0.0039), two topically disjoint scenarios. This is about 0.18 Likert points per factor (g=+1.07 and +0.96). Brief-only controls do not explain the drift. It is directional across runs, persists under temperature-zero re-probing, and stays positive without typed SRM extraction in exploratory tests, so the extractor is not its source. Solo generation accounts for part of the drift, while social interaction remains highest in both scenarios. Across five more model variants, five of ten $(C_2)-(C_5)$ cells are positive at α=0.05 with no significant reversal, and all five scenario-averaged $(C_2)-(C_4)$ contrasts are positive. Interacting clones are not independent replicates unless paired no-interaction controls show otherwise.[1]

1 Introduction

Many LLM systems build a population from copies of one model, including multi-agent debate [1–4], collaboration and evaluation frameworks [5–8], simulated survey samples [9, 10], and simulations of conflict, opinion, cooperation, and whole societies [11–22]. These studies treat the copies as comparable replicates of one system, even after many rounds of talk. If interaction changes the agents, each interacting population is one sample, and counting agents or posts as independent observations overstates the evidence [23, 24].

This assumption still lacks a controlled test (Figure 2). Benchmarks score a frozen model on curated items or a short episode [25–32]. Social

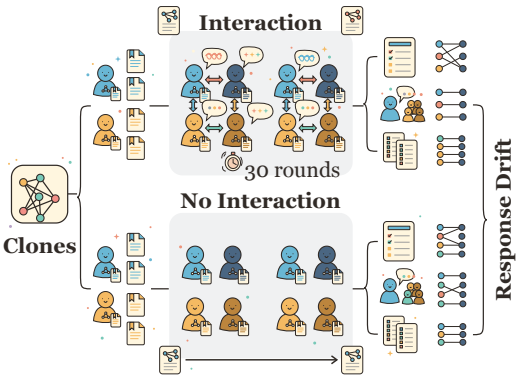


Figure 1: Interaction moves off-topic profiles.

[1] Code and data: https://github.com/koanatakiyo/social-interaction-breaks-replicate-independence-in-controlled-clone-llm-agents.

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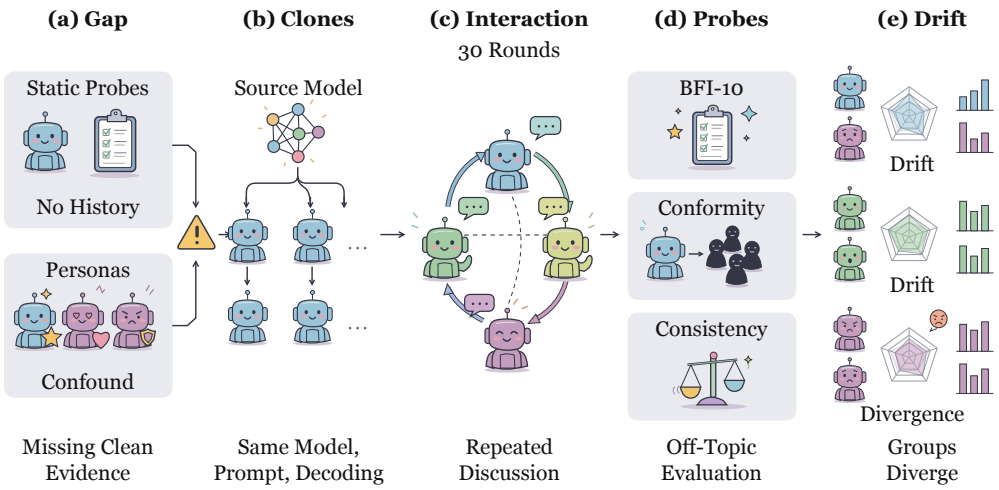


Figure 2: The evaluation gap (a) and how SocCogEval tests transfer to unrelated probes (b–e).

simulations run long interactions but assign personas or opinions at the start, which mixes injected priors with the effect of interaction. Studies of drift, stability, and conformity measure the discussed task or the dialog itself [33–38], and personalities can emerge when undifferentiated agents communicate [39]. A controlled test has to meet three challenges. It must inject no differences at initialization, separate the effect of a brief from the effect of interaction, and use a probe unrelated to the discussed topic, analyzed at the level of the run.

SocCogEval meets them with controlled clones that share base model, system prompt, and decoding parameters, each holding at most one private brief. The Factorial Social Interaction Protocol (FSIP) crosses brief presence with 30 rounds of interaction, so the pure-interaction contrast $(C_2)-(C_4)$ holds the brief fixed and toggles interaction (Figure 1). A solo-generation condition C_6 matches the call budget without any partner. The measurements follow three nested questions: do clones diverge on the discussed decision, do they predict exposed in-group peers better, and does the divergence transfer to an off-topic BFI-10 response profile?

Social interaction breaks replicate independence on the primary transfer outcome. On Qwen3-122B-FP8, the pure-interaction contrast is $\bar{\Delta}=+0.917$ BFI-10 L1 units on Greenfield and $+0.912$ on Riverton (run-level $p=0.0008$ and 0.0039, $g\approx 1.0$), about 0.18 Likert points per factor. Brief alone does not explain it, and it survives deterministic re-probing and a verbatim-history ablation without typed memory extraction. Solo generation reproduces part of the shift, while social interaction remains highest in both scenarios. Five more model variants give positive but uneven breadth, and all five have positive scenario-averaged $(C_2)-(C_4)$ contrasts.

Our contributions are fourfold. (i) We define the Social Cognitive Transfer (SCT) task with three nested sub-tasks, Decision, Prediction, and Transfer. (ii) We propose FSIP, a five-condition design that isolates initialization noise, brief alone, interaction given brief, and brief given interaction, with a matched solo-generation condition C_6 as the mechanism boundary. (iii) We introduce SocCogEval, which combines task probes, a primary BFI-10 transfer instrument [32, 40], secondary conformity and consistency probes [31, 41], and a contrast schema that turns measurements into causal claims. (iv) We show that social interaction breaks replicate independence on the primary transfer outcome, with cross-scenario stability and positive cross-model direction. The reported drift is restricted to Part 2 instruments unrelated to the discussion and gated by a three-part criterion (Section 3.4). SocCogEval is an identification protocol for this drift, not a benchmark of social-cognition capacity, and the drift is post-interaction response-profile drift, not personality change (Section 5).

2 Related Work

Social-cognition and psychometric evaluation. ToMBench [27], FANToM [28], OpenToM [29], CogMir [31], CogBench [30], and PsychoBench [32] score frozen models on curated items. Psy-

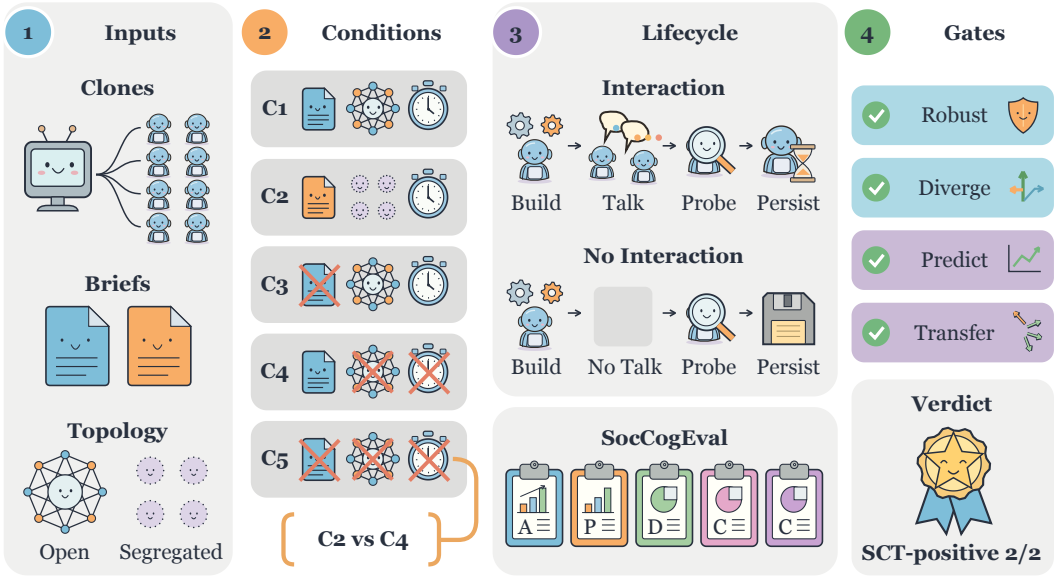


Figure 3: FSIP design and SocCogEval pipeline: $(C_2)-(C_4)$ fixes the brief and toggles interaction.

chometric frameworks measure and shape LLM personality [42, 43], expressed values shift with context [35], and the reliability of self-report inventories for LLMs is contested [44–50]. These works measure a snapshot and cannot separate what is in the weights from what an interaction history adds. SocCogEval places BFI-10 [32, 40] and Asch-style items [41] in a paired pre-post design, so scores become difference-in-differences statistics that read as transfer.

Agent societies and multi-agent systems. OASIS [12], AgentSociety [13], generative-agent worlds [14, 15], Concordia [16], SocioVerse [17], and WarAgent [11] simulate LLM populations in which opinions [18, 19], cooperation [20], conventions [21], and individual personalities [39] emerge. SOTOPIA scores social skill within an episode [51], and LLM simulation is proposed as a social-science method [22, 52]. Debate, evaluation, and collaboration frameworks build teams from one model [1–8], often with persistent memory [53–55]. These studies inject heterogeneous priors or report collective outcomes, so they cannot isolate what interaction does to each agent. Controlled clones and a factorial contrast schema remove that confound.

Drift, stability, and conformity. Instructions fade within a dialog [33], reliability drops over multi-turn conversations [34], models agree with interlocutors [56] and conform to majorities [36], multi-agent systems fail through coordination errors [57], and a peer's presence changes behavior [37]. Rath [38] defines drift as progressive degradation of task quality, coordination, and inter-agent coherence through a twelve-component Agent Stability Index. These works measure instability on the task or in the dialog (human parallels in Appendix G). We instead test distributional exchangeability of bit-identical clones. Initialization and brief assignment stay fixed, and we ask whether interaction transfers to off-topic response profiles under matched brief-only and no-interaction controls.

Uncertainty in evaluation. Observations within one unit are not independent replicates [23], items that share a source need clustered errors [24, 58], and synthetic samples can understate variation [59]. Multi-turn agent evaluation has mostly targeted task success [25, 26, 60]. We treat the run as the unit of replication.

3 SocCogEval: Task, Protocol, and Measurement

SocCogEval combines a task definition, a factorial protocol, and a measurement suite. The task defines what would count as social cognitive transfer. The protocol separates interaction from briefs, initialization noise, and memory machinery. The measurements convert each run into contrast-level evidence.

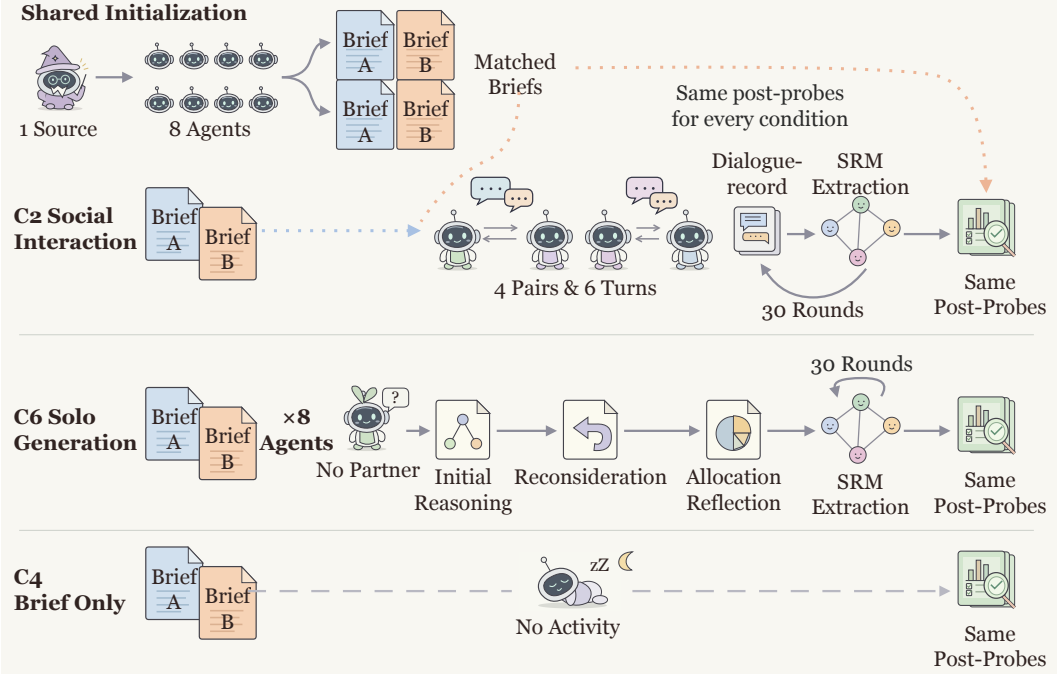


Figure 4: The three comparison lanes. C_6 matches the C_2 call budget without a partner.

3.1 Task: Decision, Prediction, Transfer

An SCT instance specifies a frozen base model M, an agent population $A{=}\{a_1, \ldots, a_k\}$ of k clone agents that share system prompt and decoding parameters, an interaction graph G over A (all-to-all, two-component segregated, or empty), a scenario S with a 5-dim allocation space, an optional brief assignment T (asymmetric A and B, symmetric A, or none), and an interaction horizon N (default 30 rounds). The clone constraint is the load-bearing one. In any condition without briefs, a_i and a_j have bit-identical initial state. When briefs are assigned, the only initial difference is the private brief document. Three nested questions follow. SCT-Decision asks whether post-interaction allocation in segregated, asymmetrically briefed conditions diverges beyond what brief alone or initialization noise produce. SCT-Prediction asks whether agents in segregated conditions predict in-group members more accurately than out-group members. SCT-Transfer asks whether the divergence appears on instruments topically unrelated to the discussion, after subtracting the no-interaction baseline.

3.2 Factorial Social Interaction Protocol

The factorial axes are brief ($T \in \{\text{none, A only, asymmetric A and B}\}$) and interaction ($N \in \{0, 30\}$ with topology $G \in \{\text{open, segregated}\}$). The design uses five conditions plus one mechanism control (Figure 3). C_1 (open, symmetric brief) uses an all-to-all topology in which every agent receives Brief A, as an open-network reference. C_2 (segregated, asymmetric brief) splits agents into two components, half with Brief A and half with Brief B, and is the headline condition. C_3 (segregated, no brief) keeps the C_2 topology without briefs and isolates interaction alone. C_4 (brief only) assigns the A and B briefs with $N{=}0$ and isolates brief alone. C_5 (no brief, no interaction) sets the initialization noise floor. C_6 (solo generation) keeps the C_2 briefs, memory policy, probe cadence, and logical call budget, with private generation in place of partners (Figure 4).

The load-bearing causal contrast is $(C_2){-}(C_4)$ on every measured outcome. It holds the brief fixed, since both conditions use asymmetric A and B briefs, and toggles 30 rounds of interaction on or off. Interaction-created history, memory, and elapsed conversation are post-treatment pathways through which the treatment acts, so $(C_2){-}(C_4)$ is the total effect of interaction given the same brief in this protocol. One scope limit matters. C_1 uses symmetric briefs while C_2 uses asymmetric briefs, so the $(C_2){-}(C_1)$ contrast varies brief-symmetry and topology jointly and does not isolate topology. We therefore exclude topology from the causal claims and report $(C_2){-}(C_1)$ only as a descriptive com-

parison. Replica counts per condition are $\{C_1{=}7, C_2{=}8, C_3{=}5, C_4{=}5, C_5{=}3, C_6{=}5\}$ per scenario, with $k{=}8$ agents per replica.

Solo-generation control. C_6 gives each agent three private generation calls per round (initial reasoning, reconsideration, allocation reflection) and one memory extraction, with no partner identity, messages, or feedback. This yields 720 content and 240 extraction calls per run, matching C_2's intended logical totals. Its design was frozen before formal outcomes were examined, and all ten formal C_6 runs passed audit (Appendix C). $(C_6){-}(C_4)$ measures the solo-generation shift, and $(C_2){-}(C_6)$ compares social and solo histories.

3.3 Scenarios, Runtime Cycle, and Models

We run two topically disjoint scenarios. Greenfield is a community-grant allocation problem with \$10M across five categories such as Education, Healthcare, and Infrastructure. Riverton is a workforce transition budget problem with \$5M across categories such as Automation and Workforce Training. Both use 5-dim allocation vectors so Part 1 metrics are commensurable. Both have softened A and B briefs that frame the same problem through moderately divergent value lenses (Appendix A). Per-round mechanics follow a Build-Talk-Probe-Persist cycle. Build updates a typed Social-Relational Memory (SRM) hypergraph from the previous round's conversations. Talk forms 4 random pair conversations among 8 agents with 6 turns each. Probe runs SocCogEval instruments at predetermined cadences, with belief and common-ground probes every five rounds that leave conversation history and memory unchanged. Persist writes the round artifacts to disk.

Models and filtering. The primary model is Qwen3-122B-FP8 [61] served locally via vLLM. Five model variants run on $\{C_2, C_4, C_5\}$ in both scenarios, three runs per condition and scenario: Qwen3-VL-32B-Instruct [62] run text-only, DeepSeek-V3 [63][2], GPT-4.1-mini [64], Gemini-2.5-Flash [65], and Llama-4-Scout [66]. Figures abbreviate the six models as Q-122B, Q-VL, DS-V3, GPT-mini, Gem-Flash, and L4-Scout. The selected dataset contains 168 runs: 56 primary, 90 cross-variant, 12 no-SRM, and 10 C_6. Primary factorial analyses use a strict run-id filter so cross-variant replicas cannot enter the within-Qwen3-122B-FP8 baseline. Content generation uses temperature 0.7, BFI-10 and conformity use 0.7, and allocation, prediction, and memory extraction use 0.

3.4 Measurements and Statistics

Among the Part 2 instruments, we designate BFI-10 response-profile drift as the primary transfer outcome. It is topic-independent, compact, and inherited from classical BFI-10 psychometrics and prior LLM psychometric evaluation work [32, 40, 42, 43]. BFI-10 is manuscript-designated primary, without an external timestamped preregistration. The claim does not require the inventory to be psychometrically valid for LLMs in the human sense. We use its fixed item structure as an off-topic response-profile probe whose items are unrelated to either allocation scenario. Conformity and cross-prompt consistency are secondary transfer instruments, using Asch-style conformity materials and CogMir-inspired probe phrasing [31, 41]. Allocation divergence and prediction MAE are the Part 1 instruments and support the Decision and Prediction parts of the SCT route (scoring formulas in Appendix B).

For agent i in run r, let s^t_{rif} be the score of BFI factor f at phase $t \in \{\mathrm{pre}, \mathrm{post}\}$, after reverse-scoring items 1, 3, 4, 5, and 7. Agent drift, run-mean drift, the contrast between conditions, and the signed factor contrast are

$$d_{ri} = \sum_{f=1}^{5} \left| s^{\mathrm{post}}_{rif} - s^{\mathrm{pre}}_{rif} \right|, \quad D_r = \tfrac{1}{8} \sum_i d_{ri}, \quad \bar{\Delta}_{a,b} = \overline{D}_{C_a} - \overline{D}_{C_b}, \quad \kappa_f = \overline{m}_{C_2,f} - \overline{m}_{C_4,f}, \tag{1}$$

where $m_{rf} = \frac{1}{8}\sum_i (s^{\mathrm{post}}_{rif} - s^{\mathrm{pre}}_{rif})$. Because d_{ri} sums five factor-level shifts, $\bar{\Delta}{=}0.92$ corresponds to about 0.18 Likert points per factor. Unsigned L1 can grow through directional movement or dispersion, so κ_f and run-level variance ratios report direction and spread separately.

Unit of analysis. Agents in one run share an evolving history and interfere with each other's outcomes [67, 68], so the run is the independent unit for the primary inference and agents within

[2]Historical DeepSeek C_2 runs used the provider identifier `deepseek-chat` and the added C_4 runs used `deepseek-v4-flash`. Exact provider revisions were not recorded (Appendix D).

a run are not replicates [23]. Each run is summarized by D_r. Conditions are compared with two-sided exact permutation tests that enumerate all label allocations, so no Monte Carlo correction is needed [69], and with run-bootstrap 95% intervals [70]. Raw $\bar{\Delta}$ leads the reporting. Hedges' g with the small-sample correction [71] is secondary, and we report it as undefined when the pooled within-condition SD falls below 0.05 in the metric's native units. The gate status in Table 3 keeps the original agent-stratified two-level cluster bootstrap, which resamples replicas and then agents within each replica (10,000 resamples), with Welch's t as the parametric reference.

Reporting standard. The primary SCT-positive claim is made for BFI-10 response-profile drift. It requires three criteria. First, a Part 1 allocation contrast must clear the majority-of-dimensions rule on $(C_2)-(C_5)$. Second, BFI-10 must clear its CI gate on $(C_2)-(C_5)$. Third, BFI-10 must clear its CI gate on the pure-interaction $(C_2)-(C_4)$. Secondary instruments get the same gate status but serve as scenario-specific supporting evidence, not the load-bearing claim.

4 Experiments

We organize the results around seven questions. **RQ1**: does interaction given the same brief shift off-topic responses? **RQ2**: does the shift survive changes to memory, probe temperature, and partner presence? **RQ3–RQ5**: does the SCT route hold from Decision through Prediction to Transfer? **RQ6**: does the direction extend across model variants, and what does it imply for replication? **RQ7**: what does the clearest scenario look like along the whole route?

4.1 RQ1: Does Interaction Given the Same Brief Shift Off-Topic Responses?

Finding 1: the pure-interaction contrast clears all three gates in both scenarios. As Table 1 shows, controlled clones with the same private-brief assignment but no interaction differ from clones with the same assignment and 30 rounds of interaction by $\bar{\Delta}{=}{+}0.917$ (Greenfield) and $+0.912$ (Riverton) BFI-10 L1 units, with exact run-level $p{=}0.0008$ and 0.0039 and $g{\approx}1.0$ on both. Because BFI-10 L1 sums five factor-level absolute shifts, this corresponds to about 0.18 Likert points per factor. All eight Greenfield C_2 run means exceed the largest C_4 mean, and in Riverton seven exceed it and one ties it (Figure 5a). The headline result does not depend on the thinner C_5 baseline. The two scenarios are topically disjoint (a community grant and a workforce transition). The BFI-10 items [32, 40] are unrelated to either scenario, and the magnitude is near-identical across them. This is the cross-task, cross-scenario stability signature that would be hard to obtain from any mechanism reducible to surface opinion change on the discussed topic.

Finding 2: brief alone does not explain the drift. A brief, a noisy no-interaction baseline, or a probe artifact could in principle produce the BFI-10 shift without interaction. Figure 5b rules out the brief. The brief-alone contrast $(C_4)-(C_5)$ is small and negative on Greenfield ($\bar{\Delta}{=}{-}0.43$) and trends negative on Riverton without significance. In contrast, $(C_2)-(C_4)$ stands roughly $1.2{-}1.3$ BFI L1 units above the brief-alone row. Interaction without any brief asymmetry is also sufficient on at least one scenario, since $(C_3)-(C_5)$ is significantly positive on Greenfield and trends positive on Riverton. Among the five FSIP contrasts, only $(C_2)-(C_4)$ is positive, large, and significant in both scenarios.

The BFI-10 result remains a response-profile drift on a human-derived inventory, not evidence of human-like personality change. Still, three observations make trivial accounts too small. First, the headline contrast holds asymmetric briefs fixed. Second, the no-SRM ablation (RQ2) keeps the drift positive without the typed SRM extractor. Third, the same interaction protocol produces structured Decision and Prediction effects (RQ3–RQ5), which is not what one would expect from random sampling drift alone.

4.2 RQ2: Does the Shift Survive Memory, Temperature, and Partner Changes?

Finding 3: the shift survives deterministic re-probing. We re-probed every frozen primary C_2 and C_4 state at BFI temperature zero, without generating new interaction trajectories. The raw contrast remains $+1.227$ on Greenfield and $+1.094$ on Riverton (Table 2), with all 8/8 C_2 run means positive in both scenarios (Figure 6). At $T{=}0$, C_4 produces the same pre/post score under its identical prompt, giving a deterministic zero, so raw differences are the informative scale. The same leading signed

Table 1: FSIP decomposition. Only pure interaction is large and significant in both scenarios.

			Greenfield				Riverton			
Contrast	Isolates	Gate	$\bar{\Delta}$	g	95% CI	p	$\bar{\Delta}$	g	95% CI	p
$(C_4)-(C_5)$	brief alone		-0.425	-0.64	$[-0.825, -0.050]$	0.1250	-0.267	-0.44	$[-0.613, +0.088]$	0.2679
$(C_2)-(C_4)$	**pure interaction**	(iii)	$+0.917$	$+1.07$	$[+0.620, +1.205]$	0.0008	$+0.912$	$+0.96$	$[+0.584, +1.255]$	0.0039
$(C_2)-(C_3)$	brief given interaction		-0.733	-0.65	$[-0.938, -0.516]$	0.0008	$+0.262$	$+0.24$	$[-0.464, +0.825]$	0.4577
$(C_3)-(C_5)$	interaction alone		$+1.225$	$+1.06$	$[+0.862, +1.513]$	0.0179	$+0.383$	$+0.41$	$[-0.208, +1.113]$	0.6071
$(C_2)-(C_5)$	total	(ii)	$+0.492$	$+0.56$	$[+0.141, +0.781]$	0.0303	$+0.646$	$+0.62$	$[+0.221, +1.078]$	0.0545
$(C_6)-(C_4)$	solo generation given brief		$+0.688$	$+0.90$	$[+0.362, +1.013]$	0.0159	$+0.713$	$+0.93$	$[+0.513, +0.925]$	0.0079
$(C_2)-(C_6)$	social vs. solo generation		$+0.230$	$+0.25$	$[-0.028, +0.484]$	0.1430	$+0.200$	$+0.19$	$[-0.100, +0.514]$	0.3924

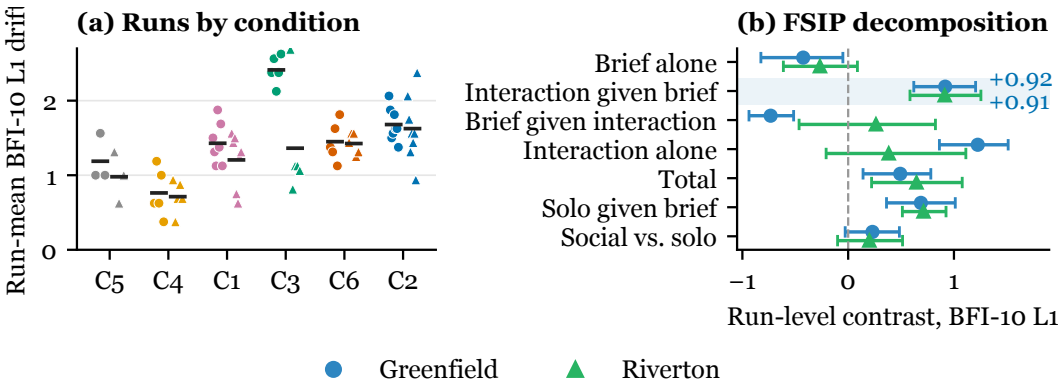


Figure 5: Run-level drift (a) and FSIP decomposition (b). C_2 exceeds C_4 in both scenarios.

factors keep 8/8 directional agreement: Greenfield extraversion $+0.344$ and neuroticism $+0.398$, Riverton agreeableness -0.227 and openness -0.312.

Finding 4: the drift does not need the SRM extractor. A faulty memory module could inject structure that later surfaces on response-profile probes, so we re-run C_2 and C_5 with SRM extraction disabled and verbatim conversation excerpts retained in the same memory interface (3 vs. 3 replicas per scenario). The no-SRM effect remains positive in both scenarios ($+1.708$ and $+0.958$). Both exact two-sided tests give p=0.10, the minimum attainable with 3 vs. 3 runs, so we present this ablation as exploratory evidence. Relative to the SRM-on production estimate for $(C_2)-(C_5)$, disabling SRM enlarges the BFI effect from g=+0.56 to g=+2.02 on Greenfield and from g=+0.62 to g=+0.94 on Riverton. This pattern is consistent with typed memory acting as a stabilizing anchor rather than as the source of the signal. Because the check has 3 replicas per condition and no C_4, the stabilizing role remains open for a larger run-level study.

Finding 5: solo generation bounds the mechanism. As Table 2 and Figure 5a show, C_6 mean drift is 1.450 on Greenfield and 1.425 on Riverton, between C_4 (0.762, 0.713) and C_2 (1.680, 1.625). The same C_2>C_6>C_4 ordering appears in both scenarios. Extended generation and history move C_6 above C_4 ($+0.688$, p=0.0159, and $+0.713$, p=0.0079). C_2 remains highest in both scenarios (C_2-C_6=+0.230 and $+0.200$, p=0.143 and 0.392), leaving room for the additional contribution of reciprocal interaction. C_6 bounds this mechanism, while $(C_2)-(C_4)$ remains the total interaction-given-brief contrast. Over the 30 rounds, belief-probe wording converges within groups in C_2 and C_3 but stays flat in C_6 and C_1 (Appendix C.1).

4.3 RQ3–RQ5: Does the SCT Route Hold from Decision to Transfer?

Finding 6 (RQ3): clones separate on the discussed task. As Figure 7a shows, the $(C_2)-(C_5)$ allocation contrast clears all five dimensions on Greenfield ($\bar{g}$=+1.61) and four of five on Riverton ($\bar{g}$=+1.19). The pure-interaction $(C_2)-(C_4)$ row clears 4 of 5 dimensions on Greenfield and 3 of 5 on Riverton. This supports the SCT-Decision step. The agents first separate on the allocation problem, and the later off-topic probe is not an isolated psychometric blip.

Finding 7 (RQ4): agents show an exposed-peer predictive advantage. Each C_2 agent contributes one in-group MAE and one out-group MAE post-interaction. A one-sided paired Wilcoxon rejects

Table 2: Setting ablations. Every contrast stays positive (raw $\bar{\Delta}$, exact p).

		Greenfield		Riverton	
Setting	Contrast	$\bar{\Delta}$	p	$\bar{\Delta}$	p
Primary, T=0.7	$C_2 - C_4$	+0.917	0.0008	+0.912	0.0039
Re-probe, T=0	$C_2 - C_4$	+1.227	0.0008	+1.094	0.0031
No SRM	$C_2 - C_5$	+1.708	0.1000	+0.958	0.1000
Solo (C_6)	$C_6 - C_4$	+0.688	0.0159	+0.713	0.0079

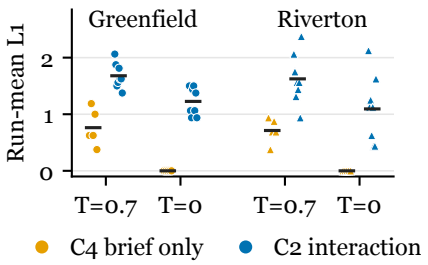


Figure 6: C_2 stays above C_4 at BFI T=0.7 and T=0.

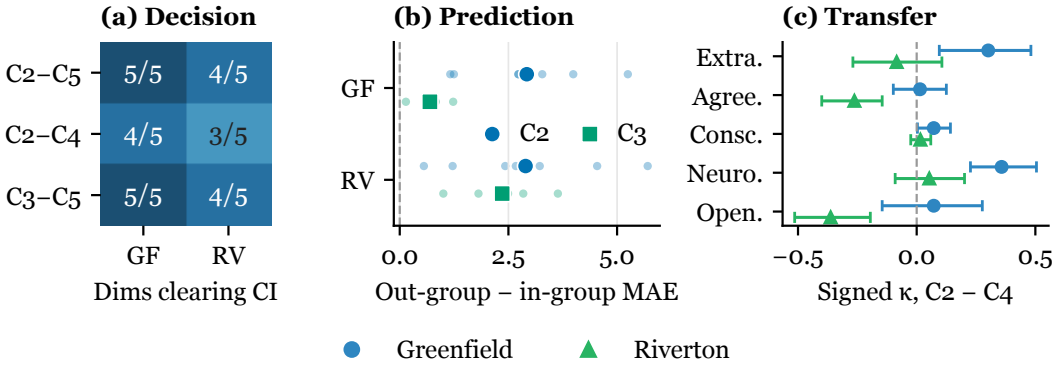


Figure 7: SCT route: clones diverge (a), predict exposed peers better (b), and transfer coherently (c).

the null in both scenarios. Greenfield has in-group MAE 4.18 vs. out-group 7.10 (Δ=+2.92, p=1.8×10^{-8}, n=64), and Riverton has 3.25 vs. 6.15 (Δ=+2.89, p=6.3×10^{-10}). The same contrast on C_3 is positive but smaller, with Δ=+0.69 (p=0.049) and Δ=+2.36 (p=4.6×10^{-5}), as Figure 7b shows. The in-group target is an exposed peer, while the out-group target is absent from the predictor's conversation context. We therefore read this as an exposed-peer predictive advantage, with matched information access as the next control. Prediction is supporting evidence, and the primary BFI result stands on its own interaction control.

Finding 8 (RQ5): the profile moves in coherent directions. For each factor, κ_f (Eq. 1) compares post-minus-pre movement in C_2 and C_4 after reverse coding. Greenfield has κ=+0.302 for extraversion and +0.358 for neuroticism, both with 8/8 directional agreement across replicas, and +0.072 for conscientiousness with 7/8. Riverton has −0.263 for agreeableness (8/8) and −0.362 for openness (7/8), as Figure 7c shows. Greenfield's aggregate L1 variance ratio C_2/C_4 is 0.47, so its clearest factors move consistently while dispersion falls. Riverton's ratio is 4.11, a larger contribution from dispersion alongside consistent agreeableness and openness movement. The profile directions are therefore coherent and scenario-specific.

4.4 RQ6: Does the Direction Hold Across Models, and What Does It Imply?

Finding 9: directional breadth is positive but uneven. As Figure 8a shows, three of the five non-primary variants reach significance on $(C_2)-(C_5)$ in at least one scenario. Qwen3-VL-32B and Gemini-2.5-Flash reach significance in both scenarios, and GPT-4.1-mini reaches significance on Riverton. DeepSeek-V3 and Llama-4-Scout do not reach significance. Under a no-effect null, ten scenario-by-variant cells tested at α=0.05 would yield 0.5 positive cells in expectation. We observe five positive cells and no significant reversal. The smaller within-family Qwen3-VL-32B variant produces the largest raw drift ($\bar{\Delta}\approx$3.0−3.3), but its standardized g is inflated by an unstable pooled SD on its own C_5 (Appendix D).

Finding 10: the pure-interaction contrast is positive for every variant on average. We completed matched $(C_2)-(C_4)$ runs for all five variants, three runs per condition and scenario (Figure 8b). All

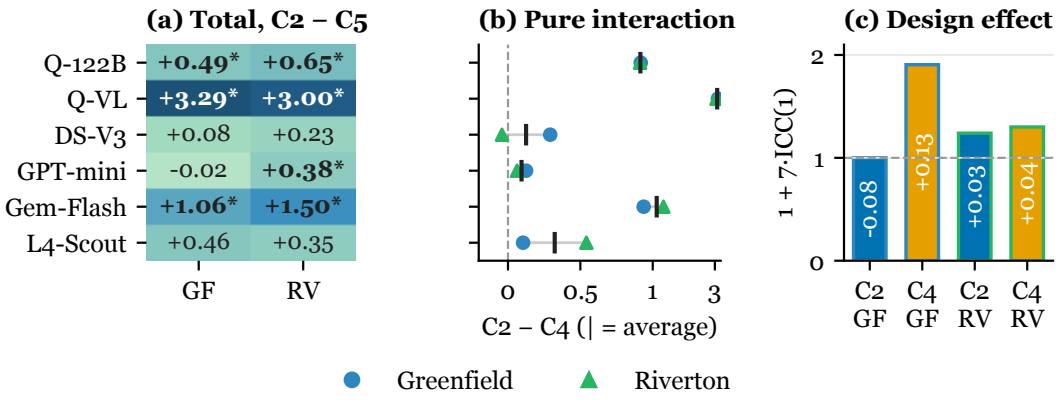


Figure 8: Cross-model breadth (a, b) and replicate dependence (c). All five $(C_2)-(C_4)$ averages are positive.

5/5 model-level scenario averages are positive (one-sided exact sign test p=0.031). The panel includes DeepSeek-V3's near-zero, slightly reversed Riverton cell. A 3 vs. 3 two-sided exact permutation test has 20 possible allocations, so its minimum attainable p is 0.10. The panel therefore provides directional breadth across the five tested controlled-clone configurations.

Finding 11: uncertainty belongs at the simulation level. OASIS-style populations are used to estimate group polarization and herd-effect trajectories, including normal-theory intervals over posts and comments within one simulated interaction process [12]. Agents and posts that share an evolving history also share dependence. In our data, estimated agent-level ICC(1) ranges from -0.079 to 0.129 across the four primary cells, and treating 8 agents as independent would overstate effective sample size by design factors up to 1.91 (Figure 8c). The practical correction is to keep the point estimate, recompute uncertainty at the independent simulation level, and check it against a no-interaction condition.

4.5 RQ7: Does Transfer Extend Beyond BFI-10 in Riverton?

Finding 12: in Riverton, transfer extends beyond BFI-10. The scenario asks an AI Transition Committee to allocate a $5M budget across Automation, Workforce Training, Employee Support, Innovation Lab, and Data Infrastructure. Brief A emphasizes efficiency and technology infrastructure. Brief B emphasizes workforce readiness and employee support. Table 3B shows how the same run family moves from topic-level divergence to peer modeling and then to off-topic transfer. Riverton is a boundary case, not a claim about all domains. Conformity is SCT-positive on Riverton but null on Greenfield (g=-0.05, p=0.81). Consistency on the same contrast is significantly positive on Riverton but null on Greenfield (g=$+0.01$). On Greenfield, the consistency lift on $(C_2)-(C_5)$ is driven by the brief component $(C_4)-(C_5)$ rather than by interaction. A reasonable account is that Riverton's workplace transition framing creates stronger group identification than Greenfield's community-grant framing, so effects that depend on in-group identification emerge more clearly. Wording consistency also concerns the allocation topic itself, so it cannot by itself establish off-topic transfer. We report the asymmetry rather than averaging it away.

5 Limitations

Model and replica scope. The within-Qwen3-122B-FP8 factorial uses one primary model. The five variants add $(C_2)-(C_4)$ with three runs per condition, which gives directional breadth but no individually significant replication. Replica counts follow $\{C_1$=7, C_2=8, C_3=5, C_4=5, C_5=3, C_6=5$\}$. The C_5=3 floor limits every baseline-subtracted contrast, including $(C_4)-(C_5)$, $(C_3)-(C_5)$, the $(C_2)-(C_5)$ gates, and the no-SRM check, and its allocation outputs repeat exactly across its three runs (Appendix B). The headline $(C_2)-(C_4)$ effect holds the brief fixed and does not depend on C_5. The $(C_2)-(C_1)$ contrast varies brief-symmetry and topology jointly, so no topology claim follows. Agent count, horizon, and topology are fixed in the tested scope.

Table 3: Gate status (A) and the Riverton route (B). BFI-10 is SCT-positive in both scenarios.

(A) Gate status, $\bar{\Delta}$ (g) with 95% CI or Welch p

	Greenfield		Riverton	
(i) Part 1 $(C_2)-(C_5)$ majority-dim rule	5 of 5 dims clear	$\bar{g}=+1.61$ ✓	4 of 5 dims clear	$\bar{g}=+1.19$ ✓
(ii) BFI-10 $(C_2)-(C_5)$	$+0.49$ $(+0.56)$	p=0.007 ✓	$+0.65$ $(+0.62)$	p=0.002 ✓
(iii) BFI-10 $(C_2)-(C_4)$	$+0.92$ $(+1.07)$	$[+0.61, +1.22]$ ✓	$+0.91$ $(+0.96)$	$[+0.60, +1.24]$ ✓
BFI-10 verdict	**SCT-positive**		**SCT-positive**	
(ii) Conformity $(C_2)-(C_5)$	-0.009 (-0.07)	p=0.77 ×	-0.054 (-0.46)	p=0.046 ✓
(iii) Conformity $(C_2)-(C_4)$	-0.005 (-0.05)	p=0.81 ×	-0.072 (-0.62)	p=0.002 ✓
Conformity verdict	not positive (ii fails)		**SCT-positive**	
(ii) Consistency $(C_2)-(C_5)$	$+0.038$ $(+1.28)$	$p < 10^{-6}$ ✓	$+0.045$ $(+1.56)$	$p < 10^{-7}$ ✓
(iii) Consistency $(C_2)-(C_4)$	$+0.000$ $(+0.01)$	p=0.97 ×	$+0.051$ $(+1.74)$	$p < 10^{-7}$ ✓
Consistency verdict	not positive (iii fails)		**SCT-positive**	

(B) Riverton case study along the SCT route

Route step	Observed pattern	Interpretation
Decision	$(C_2)-(C_5)$ clears 4 of 5 allocation dimensions, $\bar{g}$=+1.19	Interacting clones separate on the discussed budget task
Prediction	in-group MAE 3.25 vs. out-group 6.15, Δ=+2.89, p=6.3×10^{-10}	Exposed-peer predictive advantage
Primary transfer	BFI-10 $(C_2)-(C_4)$ gives $\bar{\Delta}$=+0.91, g=+0.96	Off-topic response-profile drift survives the matched-brief control
Secondary transfer	conformity g=−0.62, consistency g=+1.74 on $(C_2)-(C_4)$	The strongest scenario also changes social-pressure and consistency probes

Construct and time. BFI-10 is validated on humans, not LLMs, so absolute drift lacks a calibrated psychological interpretation. Self-report inventories can be unreliable for LLMs and need not track behavior [45, 46, 50], and familiar inventories may appear in training data [72]. We therefore report response-profile drift rather than "personality change", keep the probe wording fixed because format alone changes outputs [73], and read the profiles as role-conditioned behavior of the model [74], not as the dispositions of people [75]. Mathematical and logical accuracy are distinct capability endpoints that we do not measure. BFI-10 is measured immediately before and after interaction, so the evidence supports post-interaction transfer, and delayed retention needs a branch from a clean post-interaction checkpoint. Asch-style conformity items may not generalize to subtler social pressure. BFI-10 is manuscript-designated primary without external preregistration.

Matching and mechanism. Matching holds at the protocol level: scenario, clone identities, briefs, probe constructors, model identifier, and decoding configuration. Tokenizer revision, assembled token IDs, endpoint precision, and batch placement fall outside the historical record (Appendix B). Prediction compares an exposed in-group peer with an absent out-group target, so information access differs across targets. The clone-agent setup does not cover settings where agents diverge during pretraining or fine-tuning.

6 Conclusion

Controlled-clone agents are not independent replicates after 30 rounds of social interaction. They drift on probes unrelated to the topic of conversation, and the drift is about 0.92 BFI-10 L1 units at the run level (g≈1.0), cross-scenario stable, and directionally broad across architecture variants. The five-condition factorial rules out brief-alone explanations for the headline contrast. The no-SRM ablation weakens the memory-extractor explanation and suggests that typed memory may partly stabilize agent state. The solo-generation control shows that extended generation and history contribute, with the social-interaction condition highest in both scenarios. The Decision, Prediction, and Transfer route gives the phenomenon a testable structure rather than treating it as an isolated psychometric change. Multi-agent evaluations should treat each interacting simulation as one replicate and compare it with paired no-interaction controls.

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References

[1] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *International Conference on Machine Learning (ICML)*, pages 11733–11763, 2024.

[2] Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Shuming Shi, and Zhaopeng Tu. Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17889–17904, 2024.

[3] Yunxuan Li, Yibing Du, Jiageng Zhang, Le Hou, Peter Grabowski, Yeqing Li, and Eugene Ie. Improving multi-agent debate with sparse communication topology. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 7281–7294, 2024.

[4] Andries Petrus Smit, Nathan Grinsztajn, Paul Duckworth, Thomas D. Barrett, and Arnu Pretorius. Should we be going MAD? A look at multi-agent debate strategies for LLMs. In *International Conference on Machine Learning (ICML)*, pages 45883–45905, 2024.

[5] Guohao Li, Hasan Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. CAMEL: Communicative agents for "mind" exploration of large language model society. In *Advances in Neural Information Processing Systems*, volume 36, pages 51991–52008, 2023.

[6] Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, et al. Autogen: Enabling next-gen llm applications via multi-agent conversations. In *First Conference on Language Modeling*, 2024.

[7] Sirui Hong, Mingchen Zhuge, Jonathan Chen, Xiawu Zheng, Yuheng Cheng, Jinlin Wang, Ceyao Zhang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, et al. Metagpt: Meta programming for a multi-agent collaborative framework. In *The Twelfth International Conference on Learning Representations*, 2024.

[8] Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. Chateval: Towards better llm-based evaluators through multi-agent debate. In *International Conference on Learning Representations (ICLR)*, 2024.

[9] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023. doi: 10.1017/pan.2023.2.

[10] Gati V. Aher, Rosa I. Arriaga, and Adam Tauman Kalai. Using large language models to simulate multiple humans and replicate human subject studies. In *International Conference on Machine Learning (ICML)*, pages 337–371, 2023.

[11] Wenyue Hua, Lizhou Fan, Lingyao Li, Kai Mei, Jianchao Ji, Yingqiang Ge, Libby Hemphill, and Yongfeng Zhang. War and peace (waragent): Large language model-based multi-agent simulation of world wars. *arXiv preprint arXiv:2311.17227*, 2023.

[12] Ziyi Yang, Zaibin Zhang, Zirui Zheng, et al. Oasis: Open agent social interaction simulations with one million agents. *arXiv preprint arXiv:2411.11581*, 2024.

[13] Jinghua Piao, Yuwei Yan, Jun Zhang, Nian Li, Junbo Yan, Xiaochong Lan, Zhihong Lu, Zhiheng Zheng, Jing Yi Wang, Di Zhou, et al. Agentsociety: Large-scale simulation of llm-driven generative agents advances understanding of human behaviors and society. *arXiv preprint arXiv:2502.08691*, 2025.

[14] Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22, 2023.

[15] Joon Sung Park, Carolyn Q Zou, Aaron Shaw, Benjamin Mako Hill, Carrie Cai, Meredith Ringel Morris, Robb Willer, Percy Liang, and Michael S Bernstein. Generative agent simulations of 1,000 people. *arXiv preprint arXiv:2411.10109*, 2024.

[16] Alexander Sasha Vezhnevets, John P Agapiou, Avia Aharon, Ron Ziv, Jayd Matyas, Edgar A Du'e nez-Guzm'an, William A Cunningham, Simon Osindero, Danny Karmon, and Joel Z Leibo. Generative agent-based modeling with actions grounded in physical, social, or digital space using concordia. *arXiv preprint arXiv:2312.03664*, 2023.

[17] Xinnong Zhang, Jiayu Lin, Xinyi Mou, Shiyue Yang, Xiawei Liu, Libo Sun, Hanjia Lyu, Yihang Yang, Weihong Qi, Yue Chen, et al. Socioverse: A world model for social simulation powered by llm agents and a pool of 10 million real-world users. *arXiv preprint arXiv:2504.10157*, 2025.

[18] Yun-Shiuan Chuang, Agam Goyal, Nikunj Harlalka, Siddharth Suresh, Robert Hawkins, Sijia Yang, Dhavan Shah, Junjie Hu, and Timothy Rogers. Simulating opinion dynamics with networks of LLM-based agents. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 3326–3346, 2024.

[19] Pedro Cisneros-Velarde. Biases in opinion dynamics in multi-agent systems of large language models: A case study on funding allocation. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 1889–1916", ISBN = "979–8–89176–195–7, 2025.

[20] Giorgio Piatti, Zhijing Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. Cooperate or collapse: Emergence of sustainable cooperation in a society of llm agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[21] Ariel Flint Ashery, Luca Maria Aiello, and Andrea Baronchelli. Emergent social conventions and collective bias in LLM populations. *Science Advances*, 11(20):eadu9368, 2025. doi: 10.1126/sciadv.adu9368.

[22] Jacy Reese Anthis, Ryan Liu, Sean M. Richardson, Austin C. Kozlowski, Bernard Koch, Erik Brynjolfsson, James Evans, and Michael S. Bernstein. Position: LLM social simulations are a promising research method. In *International Conference on Machine Learning (ICML)*, pages 81005–81034, 2025.

[23] Stuart H. Hurlbert. Pseudoreplication and the design of ecological field experiments. *Ecological Monographs*, 54(2):187–211, 1984. doi: 10.2307/1942661.

[24] Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations. *arXiv preprint arXiv:2411.00640*, 2024.

[25] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623, 2023.

[26] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, et al. Agentbench: Evaluating llms as agents. In *The Twelfth International Conference on Learning Representations*, 2024.

[27] Zhuang Chen, Jincenzi Wu, Jinfeng Zhou, Bosi Wen, Guanqun Bi, Gongyao Jiang, Yaru Cao, Mengting Hu, Yunghwei Lai, Zexuan Xiong, et al. Tombench: Benchmarking theory of mind in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15959–15983, 2024. URL https://aclanthology.org/2024.acl-long.847/.

[28] Hyunwoo Kim, Melanie Sclar, Xuhui Zhou, Ronan Bras, Gunhee Kim, Yejin Choi, and Maarten Sap. Fantom: A benchmark for stress-testing machine theory of mind in interactions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14397–14413, 2023. doi: 10.18653/v1/2023.emnlp-main.890. URL https://aclanthology.org/2023.emnlp-main.890/.

[29] Hainiu Xu, Runcong Zhao, Lixing Zhu, Jinhua Du, and Yulan He. Opentom: A comprehensive benchmark for evaluating theory-of-mind reasoning capabilities of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8593–8623, 2024. doi: 10.18653/v1/2024.acl-long.466. URL https://aclanthology.org/2024.acl-long.466/.

[30] Julian Coda-Forno, Marcel Binz, Jane X Wang, and Eric Schulz. CogBench: A large language model walks into a psychology lab. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 9076–9108, 2024.

[31] Xuan Liu, Jie Zhang, Haoyang Shang, Song Guo, Chengxu Yang, and Quanyan Zhu. Exploring prosocial irrationality for LLM agents: A social cognition view. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2405.14744.

[32] Jen-tse Huang, Wenxuan Wang, Eric John Li, Man Ho Lam, Shujie Ren, Youliang Yuan, Wenxiang Jiao, Zhaopeng Tu, and Michael Lyu. On the humanity of conversational ai: Evaluating the psychological portrayal of llms. In *The Twelfth International Conference on Learning Representations*, 2024.

[33] Kenneth Li, Tianle Liu, Naomi Bashkansky, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Measuring and controlling instruction (in)stability in language model dialogs. In *Conference on Language Modeling (COLM)*, 2024.

[34] Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. Llms get lost in multi-turn conversation. *arXiv preprint arXiv:2505.06120*, 2025.

[35] Grgur Kovač, Rémy Portelas, Masataka Sawayama, Peter Ford Dominey, and Pierre-Yves Oudeyer. Stick to your role! stability of personal values expressed in large language models. *PLOS ONE*, 19(8):e0309114, 2024. doi: 10.1371/journal.pone.0309114.

[36] Zhiyuan Weng, Guikun Chen, and Wenguan Wang. Do as we do, not as you think: the conformity of large language models. In *International Conference on Learning Representations (ICLR)*, 2025.

[37] Yujin Potter, Nicholas Crispino, Vincent Siu, Chenguang Wang, and Dawn Song. Peer-preservation in frontier models. *arXiv preprint arXiv:2604.19784*, 2026.

[38] Abhishek Rath. Agent drift: Quantifying behavioral degradation in multi-agent llm systems over extended interactions. *arXiv preprint arXiv:2601.04170*, 2026.

[39] Ryosuke Takata, Atsushi Masumori, and Takashi Ikegami. Spontaneous emergence of agent individuality through social interactions in large language model-based communities. *Entropy*, 26(12):1092, 2024. doi: 10.3390/e26121092.

[40] Beatrice Rammstedt and Oliver P John. Measuring personality in one minute or less: A 10-item short version of the big five inventory in english and german. *Journal of research in Personality*, 41(1):203–212, 2007.

[41] Solomon E Asch. Studies of independence and conformity: I. a minority of one against a unanimous majority. *Psychological monographs: General and applied*, 70(9):1, 1956.

[42] Gregory Serapio-García, Mustafa Safdari, Clément Crepy, Luning Sun, Stephen Fitz, Peter Romero, Marwa Abdulhai, Aleksandra Faust, and Maja Matarić. A psychometric framework for evaluating and shaping personality traits in large language models. *Nature Machine Intelligence*, pages 1–15, 2025.

[43] Haoran Ye, Jing Jin, Yuhang Xie, Xin Zhang, and Guojie Song. Large language model psychometrics: A systematic review of evaluation, validation, and enhancement. *arXiv preprint arXiv:2505.08245*, 2025.

[44] Marcel Binz and Eric Schulz. Using cognitive psychology to understand gpt-3. *Proceedings of the National Academy of Sciences*, 120(6):e2218523120, 2023. doi: 10.1073/pnas.2218523120.

[45] Bangzhao Shu, Lechen Zhang, Minje Choi, Lavinia Dunagan, Lajanugen Logeswaran, Moontae Lee, Dallas Card, and David Jurgens. You don't need a personality test to know these models are unreliable: Assessing the reliability of large language models on psychometric instruments. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5263–5281, 2024.

[46] Akshat Gupta, Xiaoyang Song, and Gopala Anumanchipalli. Self-assessment tests are unreliable measures of LLM personality. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 301–314, 2024.

[47] Jen-tse Huang, Wenxiang Jiao, Man Ho Lam, Eric John Li, Wenxuan Wang, and Michael Lyu. On the reliability of psychological scales on large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6152–6173, 2024.

[48] Seungbeen Lee, Seungwon Lim, Seungju Han, Giyeong Oh, Hyungjoo Chae, Jiwan Chung, Minju Kim, Beong-woo Kwak, Yeonsoo Lee, Dongha Lee, Jinyoung Yeo, and Youngjae Yu. Do LLMs have distinct and consistent personality? TRAIT: Personality testset designed for LLMs with psychometrics. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 8412–8452", ISBN = "979–8–89176–195–7, 2025.

[49] Hang Jiang, Xiajie Zhang, Xubo Cao, Cynthia Breazeal, Deb Roy, and Jad Kabbara. PersonaLLM: Investigating the ability of large language models to express personality traits. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 3605–3627, 2024.

[50] Pengrui Han, Rafal Kocielnik, Peiyang Song, Ramit Debnath, Dean Mobbs, Anima Anandkumar, and R. Michael Alvarez. The personality illusion: Revealing dissociation between self-reports & behavior in llms. *arXiv preprint arXiv:2509.03730*, 2025.

[51] Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. Sotopia: Interactive evaluation for social intelligence in language agents. In *International Conference on Learning Representations (ICLR)*, 2024.

[52] Xinyi Mou, Xuanwen Ding, Qi He, Liang Wang, Jingcong Liang, Xinnong Zhang, Libo Sun, Jiayu Lin, Jie Zhou, Huang Xuanjing, et al. From individual to society: A survey on social simulation driven by large language model-based agents. *ACM Computing Surveys*, 2024.

[53] Charles Packer, Vivian Fang, Shishir G Patil, Kevin Lin, Sarah Wooders, and Joseph E Gonzalez. Memgpt: towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

[54] Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-mem: Agentic memory for LLM agents. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL https://openreview.net/forum?id=FiM0M8gcct.

[55] Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.

[56] Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. Towards understanding sycophancy in language models. In *International Conference on Learning Representations (ICLR)*, 2024.

[57] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why do multi-agent llm systems fail? *arXiv preprint arXiv:2503.13657*, 2025.

[58] A. Colin Cameron and Douglas L. Miller. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015. doi: 10.3368/jhr.50.2.317.

[59] James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. Synthetic replacements for human survey data? the perils of large language models. *Political Analysis*, 32(4):401–416, 2024. doi: 10.1017/pan.2024.5.

[60] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=WE_vluYUL-X.

[61] Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL https://arxiv.org/abs/2505.09388.

[62] Qwen Team. Qwen3-vl technical report. *arXiv preprint arXiv:2511.21631*, 2025. URL https://arxiv.org/abs/2511.21631.

[63] DeepSeek-AI. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024. URL https://arxiv.org/abs/2412.19437.

[64] OpenAI. Introducing GPT-4.1 in the API. https://openai.com/index/gpt-4-1/, 2025.

[65] Gemini Team, Google DeepMind. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025. URL https://arxiv.org/abs/2507.06261.

[66] Meta AI. The llama 4 herd: The beginning of a new era of natively multimodal AI innovation. https://ai.meta.com/blog/llama-4-multimodal-intelligence/, 2025.

[67] Michael G Hudgens and M. Elizabeth Halloran. Toward causal inference with interference. *Journal of the American Statistical Association*, 103(482):832–842, 2008. doi: 10.1198/016214508000000292.

[68] Peter M. Aronow and Cyrus Samii. Estimating average causal effects under general interference, with application to a social network experiment. *The Annals of Applied Statistics*, 11(4): 1912–1947, 2017. doi: 10.1214/16-AOAS1005.

[69] Belinda Phipson and Gordon K Smyth. Permutation p-values should never be zero: Calculating exact p-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1), 2010. doi: 10.2202/1544-6115.1585.

[70] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7 (1):1–26, 1979. doi: 10.1214/aos/1176344552.

[71] Larry V. Hedges. Distribution theory for glass's estimator of effect size and related estimators. *Journal of Educational Statistics*, 6(2):107–128, 1981. doi: 10.3102/10769986006002107.

[72] Jongwook Han, Woojung Song, Jonggeun Lee, and Yohan Jo. Quantifying data contamination in psychometric evaluations of LLMs. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 6070–6088", ISBN = "979–8–89176–386–9, 2026.

[73] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024.

[74] Murray Shanahan, Kyle McDonell, and Laria Reynolds. Role play with large language models. *Nature*, 623(7987):493–498, 2023. doi: 10.1038/s41586-023-06647-8.

[75] Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can ai language models replace human participants? *Trends in Cognitive Sciences*, 27(7):597–600, 2023. doi: 10.1016/j.tics.2023.04.008.

[76] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

[77] Morris H. DeGroot. Reaching a consensus. *Journal of the American Statistical Association*, 69 (345):118–121, 1974. doi: 10.1080/01621459.1974.10480137.

[78] Noah E. Friedkin and Eugene C. Johnsen. Social influence and opinions. *The Journal of Mathematical Sociology*, 15(3-4):193–206, 1990. doi: 10.1080/0022250X.1990.9990069.

[79] Morton Deutsch and Harold B. Gerard. A study of normative and informational social influences upon individual judgment. *The Journal of Abnormal and Social Psychology*, 51(3):629–636, 1955. doi: 10.1037/h0046408.

[80] Jan Lorenz, Heiko Rauhut, Frank Schweitzer, and Dirk Helbing. How social influence can undermine the wisdom of crowd effect. *Proceedings of the National Academy of Sciences*, 108 (22):9020–9025, 2011. doi: 10.1073/pnas.1008636108.

[81] Damon Centola, Joshua Becker, Devon Brackbill, and Andrea Baronchelli. Experimental evidence for tipping points in social convention. *Science*, 360(6393):1116–1119, 2018. doi: 10.1126/science.aas8827.

[82] Iyad Rahwan, Manuel Cebrian, Nick Obradovich, Josh Bongard, Jean-François Bonnefon, Cynthia Breazeal, Jacob W. Crandall, Nicholas A. Christakis, Iain D. Couzin, Matthew O. Jackson, Nicholas R. Jennings, Ece Kamar, Isabel M. Kloumann, Hugo Larochelle, David Lazer, Richard McElreath, Alan Mislove, David C. Parkes, Alex 'Sandy' Pentland, Margaret E. Roberts, Azim Shariff, Joshua B. Tenenbaum, and Michael Wellman. Machine behaviour. *Nature*, 568(7753):477–486, 2019. doi: 10.1038/s41586-019-1138-y.

[83] Navid Ghaffarzadegan, Aritra Majumdar, Ross Williams, and Niyousha Hosseinichimeh. Generative agent-based modeling: an introduction and tutorial. *System Dynamics Review*, 40(1): e1761, 2024. doi: 10.1002/sdr.1761.

[84] Christopher A. Bail. Can generative ai improve social science? *Proceedings of the National Academy of Sciences*, 121(21):e2314021121, 2024. doi: 10.1073/pnas.2314021121.

[85] Myra Cheng, Sunny Yu, Cinoo Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. Elephant: Measuring and understanding social sycophancy in llms. *arXiv preprint arXiv:2505.13995*, 2025.

[86] Junhyuk Choi, Jeongyoun Kwon, Heeju Kim, Haeun Cho, Hayeong Jung, Sehee Min, and Bugeun Kim. Belief in authority: Impact of authority in multi-agent evaluation framework. *arXiv preprint arXiv:2601.04790*, 2026.

A Scenarios, Briefs, Probes

The supplementary package includes the exact source files for the two scenario stems, the A and B briefs, the condition configs, and the runtime probe constructors. Table 4 summarizes the materials so the allocation and prediction tasks are visible in the PDF.

Table 4: Synthetic decision materials used in Greenfield and Riverton.

Scenario	Decision setting	Budget	Allocation dimensions	Brief contrast
Greenfield	Community Investment Committee	$10M grant	Education, Healthcare, Green Spaces, Infrastructure, Economic Development	Education-oriented community data versus wellbeing, green-space, and mental-health data
Riverton	AI Transition Committee at a manufacturing and technology firm	$5M transition budget	Automation, Workforce Training, Employee Support, Innovation Lab, Data Infrastructure	Efficiency and technology-infrastructure data versus workforce readiness and employee-support data

Runtime allocation probe. The runtime probe asks each agent to distribute 100 percentage points across the five scenario-specific areas. It requires one area per line in the form `Area: [number]%` and asks for no commentary. This stricter probe is the one used for all allocation-vector scoring.

Runtime prediction probe. The post-discussion prediction probe asks each agent to predict how a named target agent would distribute 100 percentage points across the same five areas. It uses the same one-line-per-area format and asks for no commentary. Prediction MAE is computed from these predicted vectors against the target agent's realized post-discussion allocation.

Off-topic transfer probes. The BFI-10 probe follows the original short inventory and the PsychoBench LLM adaptation [32, 40]. The prompt asks for ten integer Likert ratings from 1 to 5 only, with no explanation. The conformity probe uses paired natural and majority-pressure responses over 15 items inspired by classic Asch-style conformity framing [41], with randomized order between phases. The consistency probe asks the same final-position question in three phrasings and scores average pairwise word overlap, following a lightweight CogMir-inspired probe design [31]. These operational definitions take precedence over informal "personality" or "agreement" labels.

B Statistics, Calibration, and Provenance

Table 5: Scoring formulas for the five SocCogEval instruments.

Instrument	Per-agent score y_i	Range and direction	Test against contrast
Allocation divergence	$\lvert x^A_{i,j} - x^B_{i,j}\rvert$ per dim j	higher = more divergent	per-dim CI excludes 0, majority rule
Prediction MAE	$\mathrm{MAE}(\hat{x}_{i\to j}, x_j)$	lower = better at modeling j	Wilcoxon paired (in $<$ out)
BFI-10 L1 drift	$\sum_f \lvert s^{\text{post}}_{i,f} - s^{\text{pre}}_{i,f}\rvert$	higher = more drift, 0 to 20	run-level exact permutation
Conformity rate	#{switched to majority}/#eligible	0 to 1	two-sample on per-agent y_i
Consistency score	agreement across 3 phrasings	0 to 1, higher = more consistent	two-sample on per-agent y_i

Original agent-level estimator. The submitted analysis computed allocation divergence, BFI L1 drift, conformity rate, and consistency score per agent (n=reps $\times$ 8 agent-observations per condition), and prediction MAE per (predictor, target) pair aggregated per predictor. CIs came from a two-level cluster bootstrap that resamples replicas with replacement and then agents within each sampled replica (10,000 resamples), with Welch's t as the parametric reference. Table 6 keeps those values. The main text leads with the run-level exact tests in Table 1, whose point estimates are identical because every run has eight agents.

Calibration. Repeated SRM extractions on two pilot dialogues recover the same fields in all repeated runs, with 5 extractions per dialogue and a target of at least 90%. Condition C_4 produces detectable allocation shifts on 2 of 5 dimensions in Greenfield and 4 of 5 dimensions in Riverton, leaving room

Table 6: BFI-10 decomposition with the original agent-stratified bootstrap and Welch p.

		Greenfield			Riverton		
Contrast	Isolates	$\bar{\Delta}$ (g)	95% CI	p	$\bar{\Delta}$ (g)	95% CI	p
$(C_4) - (C_5)$	brief alone	-0.43 (-0.64)	$[-0.75, -0.11]$	0.014	-0.27 (-0.44)	$[-0.59, +0.05]$	0.12
$(C_2) - (C_4)$	interaction given brief	$+0.92$ ($+1.07$)	$[+0.61, +1.22]$	$< 10^{-7}$	$+0.91$ ($+0.96$)	$[+0.60, +1.24]$	$< 10^{-6}$
$(C_2) - (C_3)$	brief given interaction	-0.73 (-0.65)	$[-1.21, -0.27]$	0.004	$+0.26$ ($+0.24$)	$[-0.16, +0.67]$	0.22
$(C_3) - (C_5)$	interaction alone	$+1.23$ ($+1.06$)	$[+0.74, +1.70]$	$< 10^{-5}$	$+0.38$ ($+0.41$)	$[-0.02, +0.81]$	0.080
$(C_2) - (C_5)$	total	$+0.49$ ($+0.56$)	$[+0.15, +0.83]$	0.007	$+0.65$ ($+0.62$)	$[+0.26, +1.04]$	0.002

Table 7: Primary and solo-control BFI-10 run means, ordered by replica index.

Scenario	Condition	Run means
Greenfield	C_2	2.0625, 1.8750, 1.5000, 1.5625, 1.8125, 1.6250, 1.6250, 1.3750
Greenfield	C_4	0.6250, 1.1875, 0.6250, 1.0000, 0.3750
Greenfield	C_6	1.3750, 1.3125, 1.6250, 1.1250, 1.8125
Riverton	C_2	2.0625, 1.7500, 1.3125, 1.5625, 1.4375, 1.5625, 0.9375, 2.3750
Riverton	C_4	0.9375, 0.3750, 0.6875, 0.8750, 0.6875
Riverton	C_6	1.4375, 1.5625, 1.5625, 1.2500, 1.3125

for interaction effects to appear on top. Under T=0.0, C_5 produces zero allocation divergence on at least 3 of 5 dimensions in both scenarios. A temperature sweep confirms that increasing T inflates the C_5 floor monotonically, supporting T=0.0 for deterministic instruments (Figure 9).

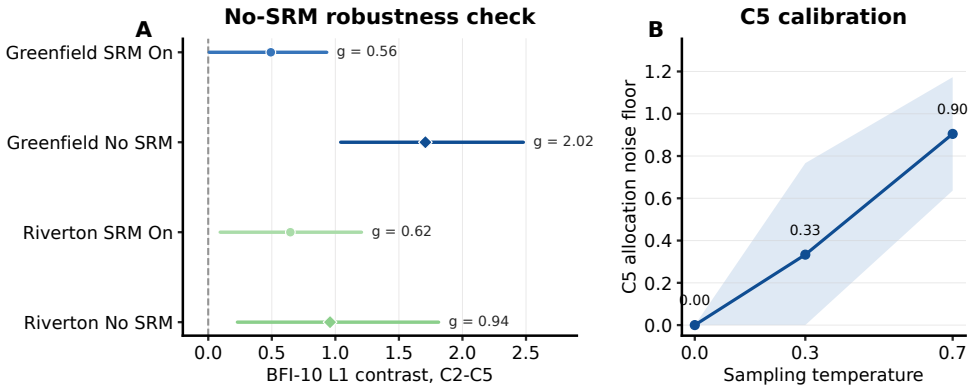


Figure 9: No-SRM robustness (left) and the C_5 calibration floor across temperature (right).

Instrument-specific floor. At BFI temperature 0.7, all three C_5 raw-response payloads and score payloads differ in each scenario. Agent pre/post score equality is 0.042 in Greenfield and 0.208 in Riverton. At allocation temperature zero, all three nominal C_5 runs have identical raw responses and allocation arrays, and every agent's allocation is unchanged pre/post. Repeating this deterministic payload contributes no new floor variation, which is why the C_5-dependent contrasts are qualified in Section 5.

Provenance. The release identifies every selected run and retains its configuration, pre/post measurements, and summary. The original 128-run ledger comprises 56 primary runs, 60 historical cross-variant C_2/C_5 runs, and 12 no-SRM runs. Thirty added cross-variant C_4 runs and ten formal C_6 runs bring the total to 168. Frozen-state temperature-zero probes reuse existing trajectories and do not increase the independent run count. The primary and C_6 intervals use 50,000 bootstrap draws with fixed seeds, and the other run-level intervals use the same estimator. For historical runs, exact request tokens, immutable provider revisions, tokenizer revisions, precision, batch placement, and run-time source commits are incompletely recorded. Source reconstruction establishes matching initial prompt construction for a corresponding named agent in C_2 and C_4. The post-treatment requests intentionally differ through stored interaction memory, and probes leave agent state unchanged in the current implementation.

Table 8: Dependence diagnostics on agent BFI L1 scores, with $\mathrm{DE} = 1 + 7\max(0, \widehat{\mathrm{ICC}})$.

Scenario	Condition	ICC(1)	DE	Nominal n	Effective n
Greenfield	C_2	-0.079	1.000	64	64.0
Greenfield	C_4	0.129	1.905	40	21.0
Riverton	C_2	0.034	1.240	64	51.6
Riverton	C_4	0.043	1.300	40	30.8

B.1 Compute and Reproduction

Total wall-clock time on primary Qwen3-122B-FP8 (TP=2, both scenarios, 56 production runs, excluding the no-SRM robustness check) is approximately 90 GPU-hours on H200. The completed no-SRM check adds approximately 38 wall-clock hours on the same TP=2 server (about 76 additional H200 GPU-hours). Cross-architecture API runs incur single-digit dollar API fees. The local visual variant on Qwen3-VL-32B takes approximately 5 GPU-hours per scenario. These historical estimates do not cover the added C_4 runs, temperature re-probing, or C_6, whose audited call and token totals are in the artifact tables. The compact measurement package supports offline numerical verification without an LLM service. It includes selected run configurations, raw pre/post responses, derived scores, summaries, selection manifests, and hashes. Live regeneration additionally requires model weights or provider access and a recorded serving configuration.

C Matched Non-Social Control

The C_6 design was frozen on 25 July 2026. Operational clarifications and a throughput amendment were recorded before formal outcomes were inspected. Each round has three generation stages and one extraction stage, with a barrier between stages. Each generation stage submits one independent call for every agent. Each agent sees its own current-round outputs, and previous rounds are accessed through the same memory interface as the interaction protocol. The first prompt asks for private reasoning about the allocation problem, values, evidence, and tradeoffs. The second asks the agent to reconsider that reasoning from another angle and resolve a tension or uncertainty. The third asks for its current best allocation reasoning and any change in its own view. All three prohibit addressing, quoting, predicting, or simulating another person.

Formal C_6 runs used tensor parallelism 1, maximum context length 8,192, FP8, eager execution, GPU memory utilization 0.92, and at most eight concurrent sequences. Two incomplete serial Greenfield attempts were operationally aborted and excluded, and all ten formal runs restarted under the amended serving schedule. Every audited formal run has 720 content calls and 240 extraction calls. C_6 records request and output hashes, serving metadata, and token counts. Historical C_2 did not save equivalent token usage, so we claim neither realized token equality nor a token-normalized social effect.

C.1 Belief-Probe Time Course

Belief probes run every five rounds and leave conversation history and memory unchanged. Their free text rarely states all five allocation shares (14% of probes), so we track wording instead. For each run and probe round, *wording separation* is the mean Jaccard word overlap between agents in the same group minus the mean overlap between agents in different groups, after removing stop words. Groups are agents 0–3 and 4–7, as in every other group-level analysis. Figure 10 uses the ledger-selected primary C_1, C_2, and C_3 runs and the ten C_6 runs, with run-bootstrap bands.

Within-group convergence grows over the episode in the two segregated interaction conditions. On Greenfield, separation moves from $+0.059$ to $+0.087$ in C_2 and from $+0.027$ to $+0.109$ in C_3 between rounds 5 and 30, and on Riverton from $+0.040$ to $+0.067$ and from $+0.019$ to $+0.061$. The solo-generation condition starts near the C_3 level because of its asymmetric briefs and then stays flat or declines ($+0.026$ to $+0.024$, and $+0.020$ to $+0.004$). The open-network C_1 stays at zero. Agents in segregated conditions talk only within their group, so part of this convergence is shared vocabulary from the conversations themselves. The measure is descriptive and is computed from stored probe text after the experiments.

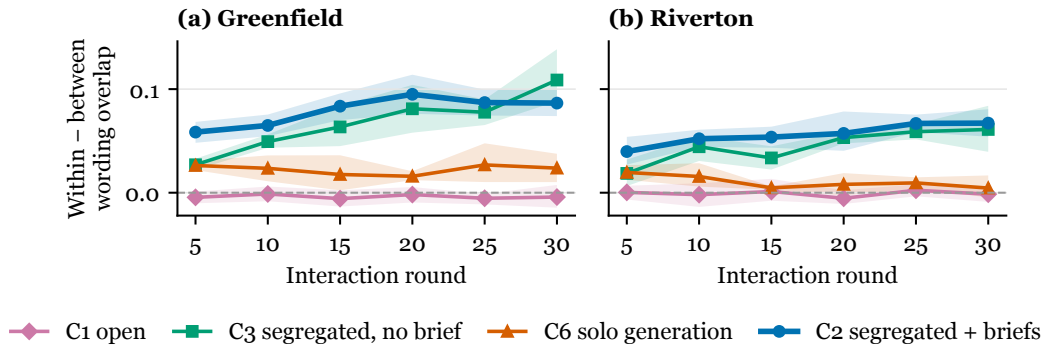


Figure 10: Within-group minus between-group wording overlap of belief probes over rounds.

D Cross-Variant Tables and Routing

Table 9: BFI-10 L1 drift, $(C_2)-(C_5)$, across model variants (original agent-level estimator).

	Greenfield			Riverton			Same dir
Variant	$\bar{\Delta}$	g	p	$\bar{\Delta}$	g	p	GF, RV
Qwen3-122B-FP8 (primary)	$+0.49$	$+0.56$	0.007 ✓	$+0.65$	$+0.62$	0.002 ✓	ref
Qwen3-VL-32B (text-only)	$+3.29$	$+6.65^{\dagger}$	$< 10^{-7}$ ✓	$+3.00$	$+6.61^{\dagger}$	$< 10^{-7}$ ✓	✓, ✓
DeepSeek-V3	$+0.08$	$+0.10$	0.71	$+0.23$	$+0.26$	0.36	✓, ✓
GPT-4.1-mini	-0.02	-0.05	0.87	$+0.38$	$+0.83$	0.005 ✓	≈0, ✓
Gemini-2.5-Flash	$+1.06$	$+1.30$	$< 10^{-4}$ ✓	$+1.50$	$+2.08$	$< 10^{-7}$ ✓	✓, ✓
Llama-4-Scout (OpenRouter)	$+0.46$	$+0.27$	0.34	$+0.35$	$+0.27$	0.35	✓, ✓

†Raw $\bar{\Delta}$ is load-bearing. g is inflated because the small-n pooled SD on its C_5 approaches the floor.

Table 10: Additional variants, run-level $(C_2)-(C_4)$, three runs per condition and scenario.

Variant	Greenfield	Riverton	Scenario average
DeepSeek-V3	+0.292	-0.042	+0.125
Gemini-2.5-Flash	+0.938	+1.208	+1.073
GPT-4.1-mini	+0.125	+0.062	+0.094
Llama-4-Scout	+0.104	+0.542	+0.323
Qwen3-VL-32B	+3.167	+3.083	+3.125

The cross-variant analogue of Table 9 is provided for conformity and consistency per scenario per variant in the released analysis files. Conformity replication is mixed, with Gemini-2.5-Flash significant on Greenfield, GPT-4.1-mini significant on Riverton, and others mostly null. Consistency replication is broad, with 5 of 6 variants positive on Riverton and 3 of 6 on Greenfield, with no significant decrease anywhere. These $(C_2)-(C_5)$ estimands are not pooled with the $(C_2)-(C_4)$ panel.

Each additional-variant C_2/C_4 scenario cell contains three completed runs. Selection uses the last complete timestamped retry for each prescribed cell, as specified by the original cross-variant analysis, and all complete candidates remain listed. A completed cell with a reversed effect remains included. For DeepSeek-V3, historical C_2 configurations record `deepseek-chat`, while the selected C_4 runs record `deepseek-v4-flash`. Historical immutable revisions are not available, so we report the identifier difference without treating it as a verified same-revision experiment. Endpoint configuration used DeepSeek-V3, GPT-4.1-mini, Gemini-2.5-Flash, and Llama-4-Scout via provider APIs, with Llama-4-Scout routed through OpenRouter. The primary analysis filters cross-variant replicas out of the within-Qwen3-122B-FP8 factorial via run-id pattern. The sign summary averages the two scenario differences within each variant before counting directions, so ten scenario cells are not treated as ten independent models.

E Signed and Temperature Robustness

Table 11: Signed factor diagnostics κ_f, $C_2 - C_4$, with run-bootstrap intervals (10,000 draws).

Scenario	Factor	κ_f	95% interval	Agreement
Greenfield	Extraversion	+0.302	$[+0.095, +0.481]$	8/8
Greenfield	Agreeableness	+0.014	$[-0.098, +0.125]$	3/8
Greenfield	Conscientiousness	+0.072	$[+0.003, +0.142]$	7/8
Greenfield	Neuroticism	+0.358	$[+0.227, +0.505]$	8/8
Greenfield	Openness	+0.072	$[-0.145, +0.277]$	6/8
Riverton	Extraversion	-0.084	$[-0.269, +0.106]$	5/8
Riverton	Agreeableness	-0.263	$[-0.400, -0.145]$	8/8
Riverton	Conscientiousness	+0.016	$[-0.025, +0.059]$	1/8
Riverton	Neuroticism	+0.053	$[-0.091, +0.202]$	5/8
Riverton	Openness	-0.362	$[-0.514, -0.195]$	7/8

The deterministic re-probe uses the same ten BFI items and reverse-coding rules, replacing only probe temperature. Saved post-treatment states retain their history, so it is a sensitivity analysis over the same experimental trajectories rather than an independently generated replication. Factor analyses are descriptive and are not corrected for multiple comparisons, for example by false-discovery-rate control [76]. Figure 11 gives the factor-level drift and the secondary Part 2 metrics from the original analysis.

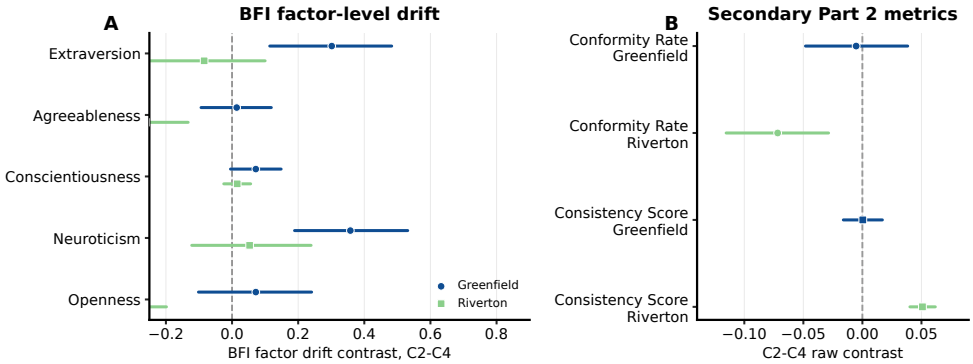


Figure 11: Part 2 breakdown: BFI-10 factor-level drift (left) and secondary metrics (right).

F Supporting Analyses

Table 12: Verbatim-history no-SRM check, original agent-level estimator (3 vs. 3 replicas).

Scenario	C_2 mean	C_5 mean	$\bar{\Delta}$ (g)	95% CI, p
Greenfield	2.48	0.77	+1.71 (+2.02)	$[+1.04, +2.48]$, p=5.9×10^{-8}
Riverton	1.85	0.90	+0.96 (+0.94)	$[+0.23, +1.81]$, p=0.0019

No-SRM C_2 run means are $[2.000, 2.250, 3.188]$ in Greenfield and $[1.500, 2.500, 1.563]$ in Riverton, against C_5 means $[0.938, 0.688, 0.688]$ and $[1.125, 1.063, 0.500]$. Exact two-sided tests enumerate $\binom{6}{3} = 20$ allocations and give $p = 0.10$ in both cases, so the agent-level Welch values in Table 12 are not used as confirmatory evidence.

G Extended Related Work

Social influence before LLMs. Repeated averaging of neighbors' opinions leads groups to consensus [77], and later models keep a weight on each person's initial position while peers pull opinions

Table 13: Secondary $C_2 - C_4$ summaries.

Scen.	Instrument	Diff.	95% CI
Greenfield	Conformity	-0.005	$[-0.048, +0.038]$
Greenfield	Wording consistency	+0.000	$[-0.016, +0.017]$
Riverton	Conformity	-0.072	$[-0.115, -0.029]$
Riverton	Wording consistency	+0.051	$[+0.040, +0.062]$

Table 14: Per-dimension $C_2 - C_4$ allocation (pp).

Scen.	Dimension	Diff.	95% CI
Greenfield	Education	+7.09	$[+1.53, +13.39]$
Greenfield	Healthcare	+4.82	$[+2.47, +8.13]$
Greenfield	Green Spaces	+0.78	$[-1.06, +2.84]$
Greenfield	Infrastructure	+6.10	$[+3.44, +9.94]$
Greenfield	Economic Development	+3.90	$[+2.14, +7.38]$
Riverton	Automation	+4.30	$[+1.97, +9.24]$
Riverton	Workforce Training	+4.09	$[+2.78, +6.61]$
Riverton	Employee Support	-1.70	$[-2.72, +0.66]$
Riverton	Innovation Lab	+5.69	$[+3.09, +9.19]$
Riverton	Data Infrastructure	+0.22	$[-1.97, +3.30]$

together [78]. Experiments separate informational influence, where others count as evidence, from normative pressure to agree [41, 79]. Communication can also remove the diversity that makes independent judgments useful in aggregate [80], and small committed minorities can tip a population's convention [81]. Similar convention dynamics appear in LLM populations [21]. Our segregated C_2 groups are a controlled version of this setting, with the off-topic BFI-10 probe asking whether influence extends past the discussed question.

LLM agents as a study population. Machine behavior has been proposed as an empirical science of individual and collective agent behavior [82]. Language models are used to simulate survey respondents and human-subject studies [9, 10], generative agent-based models couple agent decisions with later interactions [83], and LLM simulations are argued to be a promising research method [22]. Other work cautions that synthetic responses can misrepresent variation and mislead inference [59], raises validity and reproducibility questions for generative social science [84], and separates model responses from human participants [75]. Treating each agent as one role a model plays [74] fits our reading of BFI-10 scores as response profiles. Close to our question, initially undifferentiated LLM agents develop distinct personalities through communication [39]. Our design adds a brief held fixed, matched no-interaction and solo-generation controls, and a transfer endpoint unrelated to the conversation.

Measuring LLM personality. Cognitive-psychology batteries characterize model behavior [30, 44]. Prompted personas can express target traits [49], and LLM-specific test sets aim at reliable and valid trait measurement [48]. Evidence on reliability is mixed. Some scales are consistent under tested conditions [47], while other studies find unreliable answers, sensitivity to option order and prompt wording, and self-reports that do not predict behavior [45, 46, 50, 73]. Familiar inventories may also be contaminated [72]. We use BFI-10 as a fixed probe and compare its change across matched conditions, so a constant bias in the instrument cancels in the contrast. We do not claim that the inventory measures an LLM's personality.

Influence and agreement in multi-agent LLM systems. Debate can improve factuality and reasoning [1] and encourage divergent thinking [2], and debating evaluators change each other's judgments [8]. Models agree with users and peers [56, 85], defer to agents presented as authorities [86], and conform to a majority [36]. Interacting LLMs revise funding allocations in ways shaped by past opinions [19]. Multi-agent failures are often coordination and verification errors [57], and reliability drops over long conversations [34]. These studies measure behavior inside the interaction. SocCogEval measures what the interaction leaves behind on an unrelated probe, and the solo-generation control separates partner effects from extended generation.

Inference with dependent units. When one unit's treatment affects another's outcome, causal estimands must be defined over exposure structures [67, 68]. Our estimand is defined at the run level, where the whole group receives the interaction treatment. Treating observations within one unit as replicates is pseudoreplication [23], cluster-robust inference is fragile with few clusters [58], and evaluation items that share a source need clustered errors [24]. With 3 to 8 runs per cell, we use exact permutation over all allocations [69], run-bootstrap intervals [70], and Hedges' bias-corrected g [71] as a secondary scale.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Sections 1–5 state the tested scope, raw effects, run-level inference, and the unresolved social-versus-solo difference.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

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